Thomas Puskas

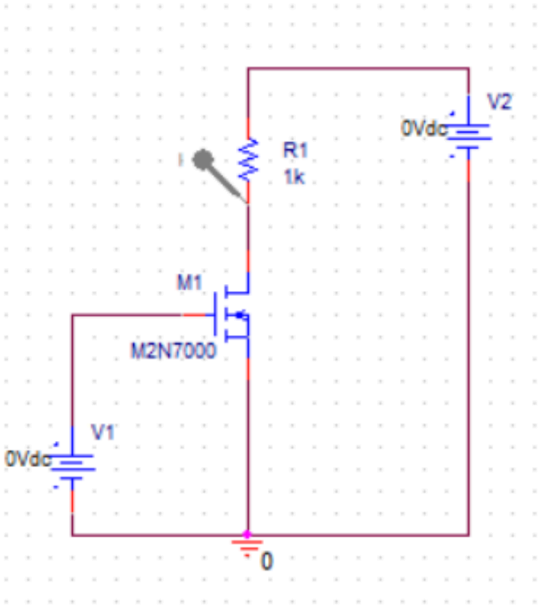
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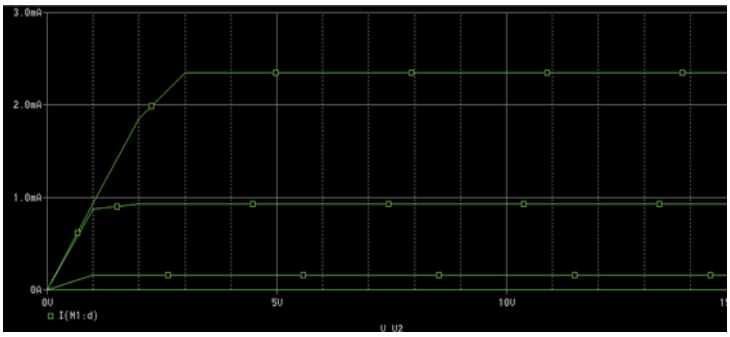
Experiment 10: MOSFET Common Source Amplifier

Objective: Determine through experiments the characteristics of the drain of a MOSFET and measuring the voltage gain of a common source amplifier MOSFET. We also want to study the effect and purpose of source degeneration.

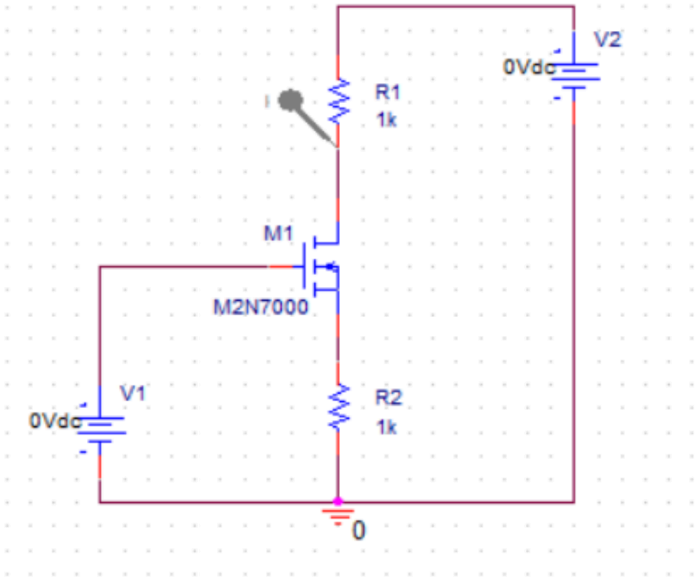
Pre-Lab:

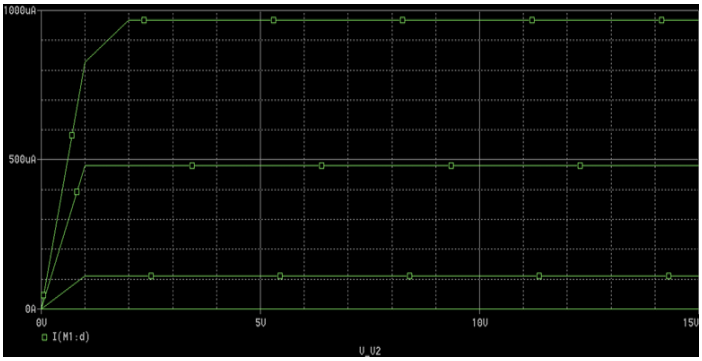
Experiment 10.1



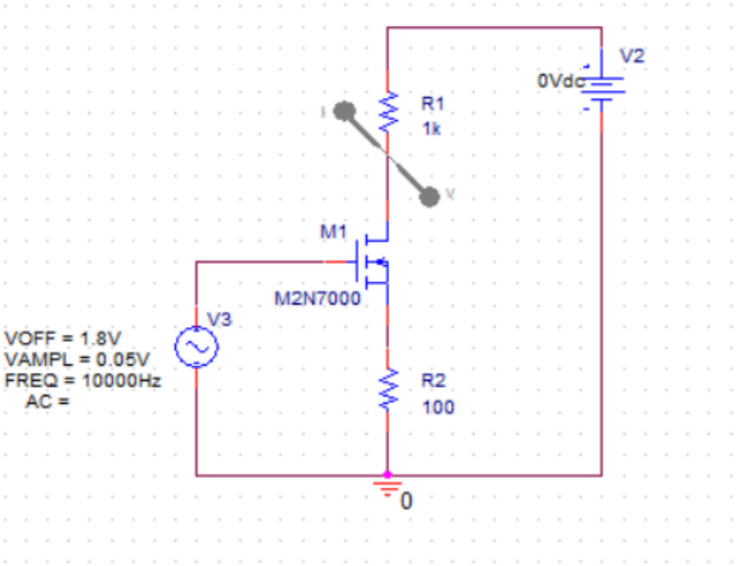


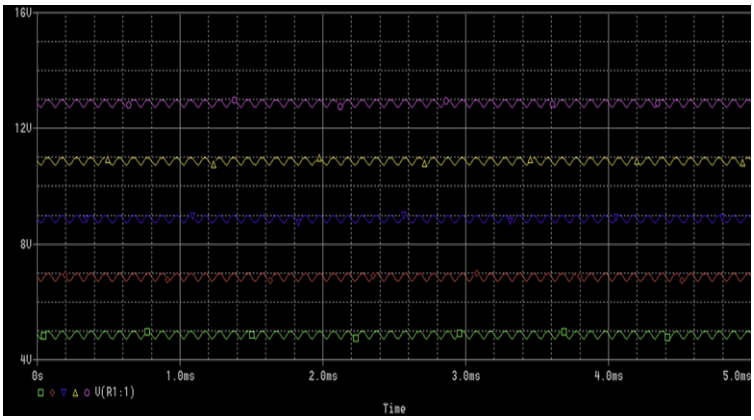
Experiment 10.2





Experiment 10.3





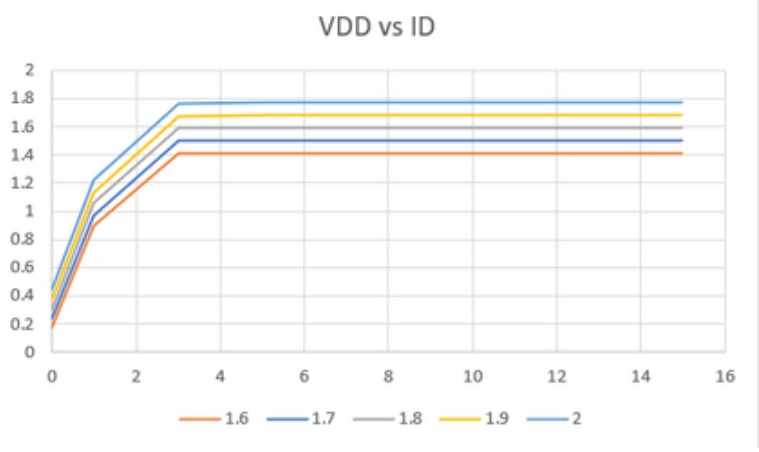
Lab:

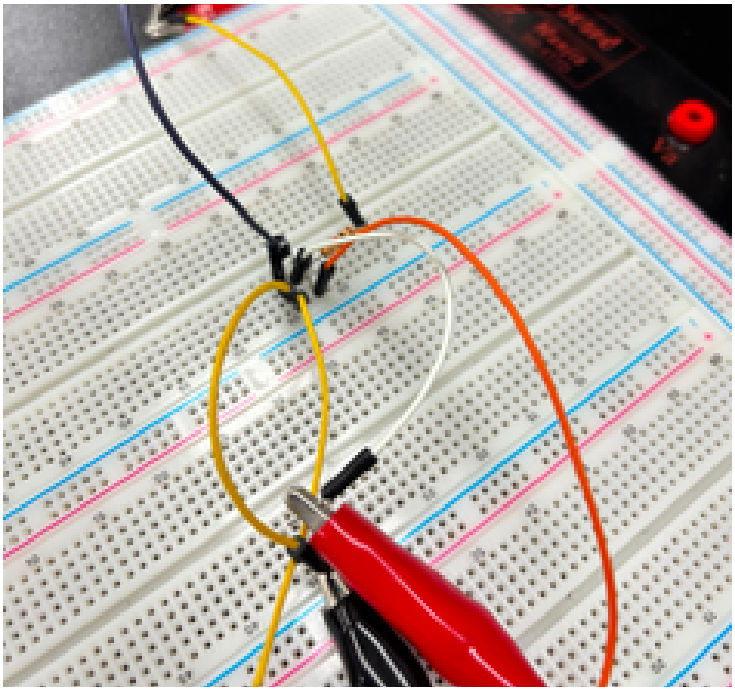
Experiment 10.1:

1. First we will set up the circuit as shown in Figure with , .
2. And then repeat the steps to fill up the table:

[r

[V]	0	1	3	5	7	9
[V]						
1.6	0.176	0.90	1.41	1.41	1.41	1.41
1.7	0.234	0.97	1.50	1.50	1.50	1.50
1.8	0.300	1.059	1.59	1.59	1.59	1.59
1.9	0.371	1.13	1.67	1.68	1.68	1.68
2	0.445	1.22	1.76	1.77	1.77	1.77





Circuit:

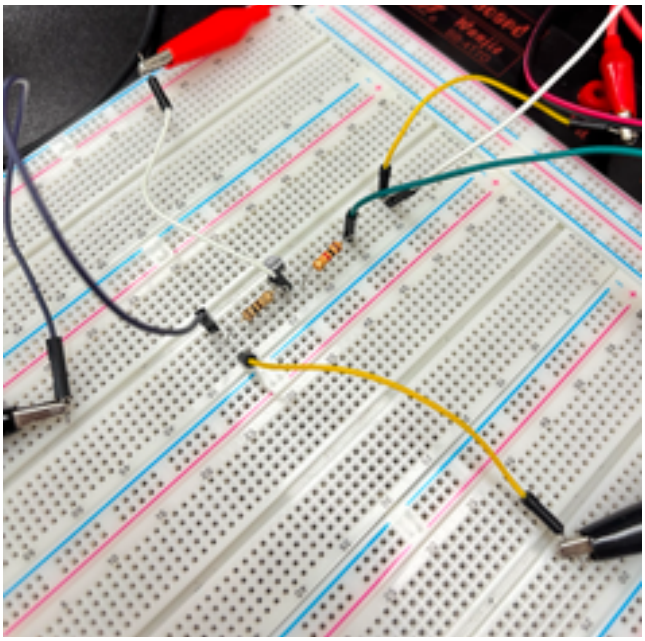




Experiment 10.2

1. Now we set up circuit with R1 as 1kOhm and R2 as 100Ohm as shown in the figure.
2. And then the values are recorded.

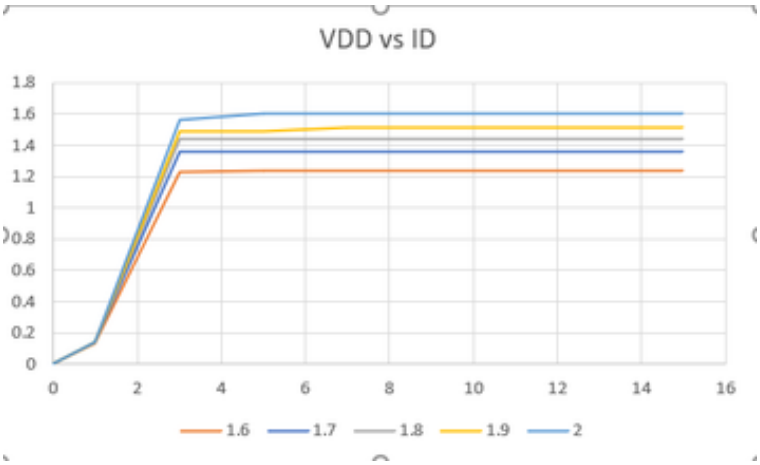
					[r
[V]	0	1	3	5	.
[V]					
1.6	0	0.133	1.23	1.24	1.
1.7	0	0.138	1.36	1.36	1.
1.8	0	0.136	1.44	1.44	1.
1.9	0	0.142	1.49	1.49	1.
2	0	0.140	1.56	1.60	1.



Circuit:

VGG at 1.7V and VDD at 15V



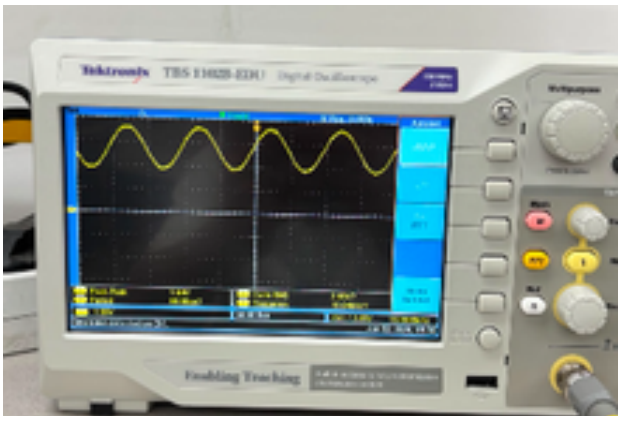


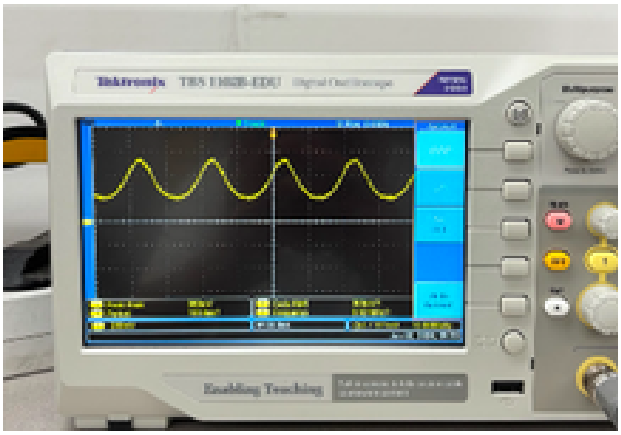
In the case of source degeneration it is important to note that ID is saturated more quickly and more flat. The source resistor causes negative feedback and therefore when ID increases the source voltage also increases and then the effective VGS is reduced.

Experiment 10.3

1. Now we continue with the same circuit as shown in figure below but then replace VGG with the sine signal generation.
2. Then we will apply VDD to 5V and set the function generator (Vin) offset to 1.8V and amplitude to 0.05V.
3. The frequency is 10kHz
4. VDD is increased and the results are recorded.
5. We then disconnect R2 and repeat the steps and the results are recorded.

	[mV]			
[V]	with	without	with	without
5	360	336	3.60	3.36
7	400	1220	4.00	12.20
9	480	1840	4.80	18.40
11	416	2160	4.16	21.60
13	420	2160	4.20	21.60





The main differences between the gain without R2 compared to with it is because the voltage gain was higher in a significant way when R2 was removed. Because of this we know with R2 that the gain remained constant but when R2 was removed, it increased significantly. This occurs because R2 produces negative feedback and so when the drain current increases it causes the source voltage to also increase.